

Open-source Toolkit for Seismic Vulnerability Assessment

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Abstract

This study develops an open-source numerical modeling toolkit and library for assessing the seismic vulnerability of building structures. It leverages an object-oriented procedure via OpenSeesPy software to create numerical models for a wide range of building classes using multi-degree-of-freedom oscillators. These models reliably capture the dynamic behavior of various structural types, and are applicable for risk analysis of large building portfolios. The toolkit features various analytical capabilities, including linear analyses (e.g., gravity and modal analyses), nonlinear static analyses (such as static and cyclic pushover analyses), and dynamic nonlinear time-history analyses (i.e., nonlinear time-history analyses), enabling a detailed evaluation of seismic performance. Furthermore, the toolkit is enhanced with a component for ground-motion selection and simulation to enhance its utility in analysis tailored for site-specific seismic hazards or desired ground-motion characteristics. The toolkit also facilitates the generation of probabilistic demand-intensity models through robust regression of cloud analysis results, linking seismic demand to building response, and derives fragility functions to estimate the probability of exceeding various damage states under increasing levels of ground shaking. Vulnerability functions to quantify the overall expected losses (and eventually risk) for the building classes are generated by the toolkit as well. The toolkit offers a flexible library for seismic vulnerability assessment, and its open-source nature allows for transparency, reproducibility, and ongoing development. The tool is maintained by the Global Earthquake Model (GEM) Foundation on GitHub and benefits from continuous updates and contributions from researchers worldwide.